Problem 1. Show that additive inverses are *unique*. (That is, if $a \in \mathbb{R}$, then there is only one $b \in \mathbb{R}$ such that $a + b = 0$.)

Proof. Let $a \in \mathbb{R}$. By the additive inverse axiom, we know that there exists an element $b \in \mathbb{R}$ such that

$$a + b = 0.$$

Assume that there exists an element $c \in \mathbb{R}$ such that $a + c = 0$ also. (I.e., c is an additive inverse of a.) By symmetry of equality, $0 = a + c$ so by the transitivity of equality,

$$a + b = a + c.$$

By the Cancellation Theorem, this implies that $b = c$. Therefore all additive inverses of a are actually the same element. In other words, additive inverses are unique. □

Problem 2. Let $a, b, c \in \mathbb{R}$. If $a > b$ and $c > 0$, then $ac > bc$.

Proof. Let $a, b, c \in \mathbb{R}$, and assume that $a > b$ and $c > 0$. We will show that $ac > bc$.

By our definitions following the order axioms, we know that $a - b \in \mathbb{R}^+$. By a theorem from class, since $c > 0$, therefore $c \in \mathbb{R}^+$. By the positive closure axiom, $(a - b)c \in \mathbb{R}^+$. By the distribution axiom, $(a-b)c = ac-bc$. Therefore, by the substitution axiom, $ac-bc \in \mathbb{R}^+$, which means precisely that $ac > bc$. This completes the proof. □

Problem 3. Either prove the following or provide a counterexample to show that it is false:
For all $a, b \in \mathbb{Z}$, if ab is even, then a is even or b is even.

Proof. We will prove this statement by contrapositive, which is:

For all $a, b \in \mathbb{Z}$, if neither a nor b are even, then ab is not even.

By a theorem from class, an integer is not even precisely when it is odd (and vice-versa). We assume that $a, b \in \mathbb{Z}$ and that they are both odd. Thus $a = 2k + 1$ and $b = 2\ell + 1$ for some $k, \ell \in \mathbb{Z}$. Then

$$ab = (2k + 1)(2\ell + 1) = 4k\ell + 2\ell + 2k + 1 = 2(2k\ell + \ell + k) + 1.$$

Since $2k\ell + \ell + k \in \mathbb{Z}$, this shows that ab is odd. Again by the theorem from class, this means that ab is not even. This completes the proof. □

Problem 4. Prove the following theorem:
Let $n \in \mathbb{Z}$. If $n^2 - 1$ is divisible by 2, then $n^2 - 1$ is divisible by 4.

Proof. Let $n \in \mathbb{Z}$. Assume that 2 divides $n^2 - 1$. We will show that 4 divides $n^2 - 1$.

Since 2 divides $n^2 - 1$, this means precisely that $n^2 - 1 = 2m$ for some $m \in \mathbb{Z}$. Observe that $n^2 - 1 = (n - 1)(n + 1)$, so $(n - 1)(n + 1) = 2m$. Since $(n - 1)(n + 1)$ is even, the previous problem shows that either $n - 1$ or $n + 1$ is even.

If $n - 1$ is even, then $n - 1 = 2k$ for some $k \in \mathbb{Z}$ and thus $n + 1 = 2k + 2 = 2(k + 1)$. If instead $n + 1$ is even, then $n + 1 = 2\ell$ for some $\ell \in \mathbb{Z}$ and thus $n - 1 = 2\ell - 2 = 2(\ell - 1)$. in either case, *both* $n - 1$ and $n + 1$ are even. Therefore, without loss of generality,

$$n^2 - 1 = (n - 1)(n + 1) = (2k)(2k + 2) = 4(k^2 + k).$$

Since $k^2 + k \in \mathbb{Z}$, this shows that 4 divides $n^2 - 1$, which completes the proof. □

Problem 5. Recall that a real number x is *rational* if we can write $x = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. We say that x is *irrational* if it is impossible to write x in this way. Either prove the following statement or provide a counterexample to show that it is false:

Let $x \in \mathbb{R}$ with $x \geqslant 0$. If x is irrational, then $\sqrt{x}$ is irrational.

Proof. We will prove this by contrapositive. Let $x \in \mathbb{R}$ with $x \geqslant 0$ and assume that $\sqrt{x}$ is rational. Therefore $\sqrt{x} = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. Thus $x = \sqrt{x}^2 = \left(\frac{p}{q}\right)^2 = \frac{p^2}{q^2}$. The integers are closed under multiplication, so $p^2, q^2 \in \mathbb{Z}$. Hence, to show that x is rational, we just need to prove that $q^2 \neq 0$.

Suppose for contradiction that $q^2 = 0$. Since $q \neq 0$, it has a multiplicative inverse, q^{-1}. Then $q = q^{-1}(q^2) = q^{-1}0 = 0$. But this contradicts the fact that $q \neq 0$.

Therefore, we have written $x = \frac{p^2}{q^2}$ as a fraction of two integers, with the denominator nonzero. This completes the proof by contrapositive. □

Problem 6. Prove or provide a counterexample to the following two statements:

1. The sum of two rational numbers is rational.

   Solution. This statement is true. We provide a proof.

   Proof. Suppose that $x, y \in \mathbb{Q}$. Then there exist integers p, q, r, s with $q, s \neq 0$ such that $x = \frac{p}{q}$ and $y = \frac{r}{s}$. Now
   $$x + y = \frac{p}{q} + \frac{r}{s} = \frac{ps}{qs} + \frac{rq}{qs} = \frac{ps + rq}{qs}.$$
   Note that $ps + rq, qs \in \mathbb{Z}$ by closure of the integers. Hence, to complete our proof that $x + y \in \mathbb{Q}$, we just need to show that $qs \neq 0$.

   We assumed that $q \neq 0$ and $s \neq 0$. Suppose for contradiction that $qs = 0$. Then q has a multiplicative inverse, q^{-1} so $s = q^{-1}(qs) = q^{-1}0 = 0$. But this contradicts the fact that $s \neq 0$.

   Hence, $qs \neq 0$, which completes our proof. □

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2. The sum of two irrational numbers is irrational.

   Solution. This statement is false. We proved in class that $\sqrt{2}$ is irrational. Since, by the inverse axiom, the additive inverse of a rational number is rational, this implies that $-\sqrt{2}$ is also irrational. Now notice that $\sqrt{2} + (-\sqrt{2}) = 0$ and 0 is rational since we can write it as $0/1$. This gives a counterexample to the statement above. □

Problem 7. Either prove the following or provide a counterexample to show that it is false:

Let $n \in \mathbb{Z}$. If n is odd, then $n^2 \equiv 1 \pmod 8$.

Proof. Let $n \in \mathbb{Z}$, and assume that n is odd. Therefore $n = 2k + 1$ for some $k \in \mathbb{Z}$. We will show that $n^2 \equiv 1 \pmod 8$. Notice that

$$\begin{aligned} n^2 - 1 &= (2k + 1)^2 - 1 \\ &= (4k^2 + 4k + 1) - 1 \\ &= 4(k^2 + k) = 4k(k + 1). \end{aligned}$$

We know by a theorem from class that every integer is either or not (and not both). This means that k is either even or not. We will consider each case separately.

Case I. k is even.
If k is even, then $k = 2m$ for some $m \in \mathbb{Z}$, so

$$n^2 - 1 = 4k(k+1) = 4(2m)(2m+1) = 8m(2m+1).$$

Case II. k is odd.
If k is odd, then $k = 2m+1$ for some $m \in \mathbb{Z}$, so

$$n^2 - 1 = 4k(k+1) = 4(2m+1)(2m+2) = 8(2m+1)(m+1).$$

Notice that in either case that 8 divides $n^2 - 1$. This means precisely that $n^2 \equiv 1 \pmod 8$, which completes the proof. □

Problem 8. Complete Exercise 20 on pages 129 and 130 of the textbook (*Mathematical Reasoning* by Sundstrom).

1. **Proposition.** For each real number x, if x is irrational and m is an integer, then mx is irrational.

   Solution. This statement is false. Let x be any irrational number and $m = 0$. Then $mx = 0 = \frac{0}{1} \in \mathbb{Q}$. Therefore the statement is not true in general. (In fact, this is the only m value that is a problem.) □

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2. **Proposition.** For all real numbers x and y, if x is irrational and y is rational, then $x + y$ is irrational.

   Solution. True proposition and beautiful proof. It uses full sentences, lets the reader know about proof structure (e.g. "We will use a proof by contradiction.") Every variable, term, etc. is clearly stated and explained. Perfect. □

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3. **Proposition.** For each real number x, $x(1-x) \leqslant \frac{1}{4}$.

   Solution. The statement is true, but the given proof has a humongous major logical flaw, so the proof is not correct. There is also a small arithmetic error (since $4 \cdot 3(1-3) = -24$), but this arithmetic error is *not* what makes the proof incorrect. If you correct the arithmetic error, the given proof still has the huge flaw.

   Overall, the writing style and (to a certain extent) the structure is fine. They have correctly stated their assumption for a proof by contradiction as "There exists an $x \in \mathbb{R}$ such that $x(1-x) > \frac{1}{4}$". However, if this is true, that does not imply that $3(1-3) > \frac{1}{4}$ (all they know is that there exists *some* x such that $x(1-x) > \frac{1}{4}$, that does not mean that 3 has this property).

   In other words, they have found a contradiction to "For all $x \in \mathbb{R}$, $x(1-x) > \frac{1}{4}$", but this was not what they needed to contradict. Here's a better proof (we'll use a direct proof):

   Proof. Let $x \in \mathbb{R}$. We will show that $x(1-x) \leqslant \frac{1}{4}$. Observe that $2x - 1 \in \mathbb{R}$, and therefore $0 \leqslant (2x-1)^2$. Thus

   $$\begin{aligned} 0 &\leqslant (2x-1)^2 \\ 0 &\leqslant 4x^2 - 4x + 1 \\ 4x - 4x^2 &\leqslant 1 \\ x(1-x) &\leqslant \frac{1}{4} \end{aligned}$$

   which is the desired result. This completes the proof. □

Problem 9. Do not turn these in, but I recommend doing the progress checks for sections 3.3 and 3.4 in the textbook.